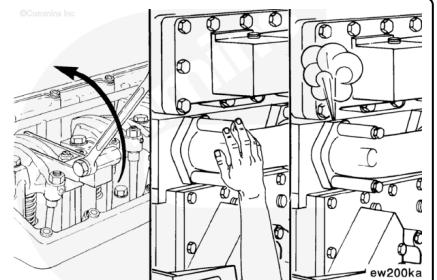


Trainings ▾

General Information

⚠ WARNING ⚠

Due to the stroke of the injectors on M series engines, do not use the mechanical method to perform a cylinder cutout test. When an injector on an engine is mechanically shorted (barred), the push rods can be thrown free, causing personal injury. Refer to the Automated Cylinder Performance Test to perform the test on M series engines.



The following methods can be used to find the cylinder(s) causing misfire/white smoke:

- Exhaust manifold temperature
- Tilting back exhaust manifold
- Automated cylinder performance test.

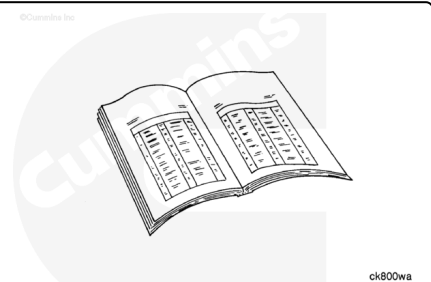
Automated Cylinder Performance Test

Do **not** run the automated cylinder performance test on engines that are smoking at idle.

Perform the automated cylinder performance test at least twice.

Refer to the appropriate electronic service tool manual.

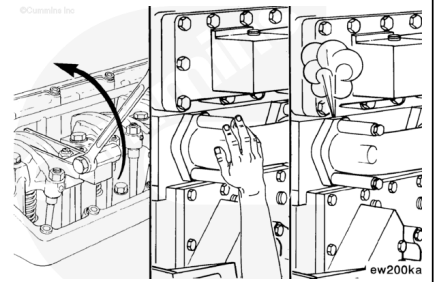
Minimum cylinder contribution is 71 percent.



Cylinder Cutout Test

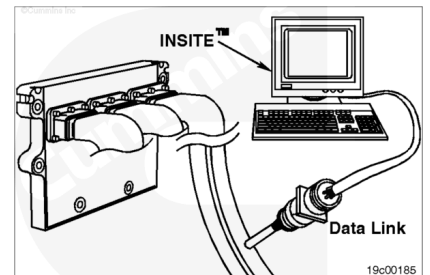
The following methods can be used to find the cylinder(s) causing misfire/white smoke:

- Exhaust manifold temperature
- Tilting back exhaust manifold
- Automated cylinder performance test.



Cut out each cylinder while watching the exhaust smoke. Refer to the appropriate electronic service tool manual.

Replace the injector in the cylinder that causes a reduction in smoke when cut out.



Exhaust Manifold Temperature Test

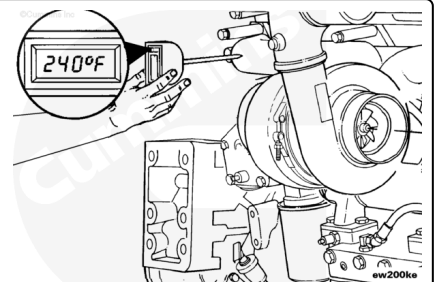
To locate a smoking cylinder, use a pyrometer or infrared temperature gun to measure the exhaust manifold surface temperature at each cylinder port.

The maximum exhaust manifold surface temperature at idle is 143°C [290°F].

A cylinder is smoking if:

- The exhaust manifold surface temperature at that exhaust port exceeds the above specification.
- The exhaust manifold surface temperature at that port is significantly higher than that for the other cylinder ports.

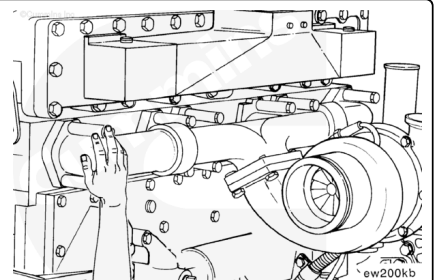
If this test is inconclusive, tilt back the exhaust manifold to locate the smoking cylinder.



⚠ WARNING ⚠

To reduce the possibility of personal injury, do not place hand on the exhaust manifold if the engine is hot or emitting thick black smoke.

An alternative method is to place your hand on the exhaust manifold at each cylinder port. A smoking cylinder will feel hot.



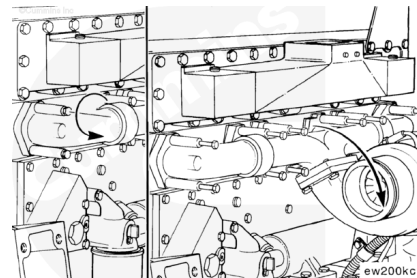
Replace the injector in the cylinder that is producing the smoke.

Refer to Procedure 006-026 in Section 6.

(/qs3/pubsys2/xml/en/procedures/35/35-006-026-tr.html)

Exhaust Manifold Tilt-back Method

Loosen the exhaust manifold capscrews, and tilt the exhaust manifold back.



Idle the engine.

Observe the smoking cylinder.

Replace the injector in the cylinder that is producing the smoke.

Refer to Procedure 006-026 in Section 6.

(/qs3/pubsys2/xml/en/procedures/35/35-006-026-tr.html)

